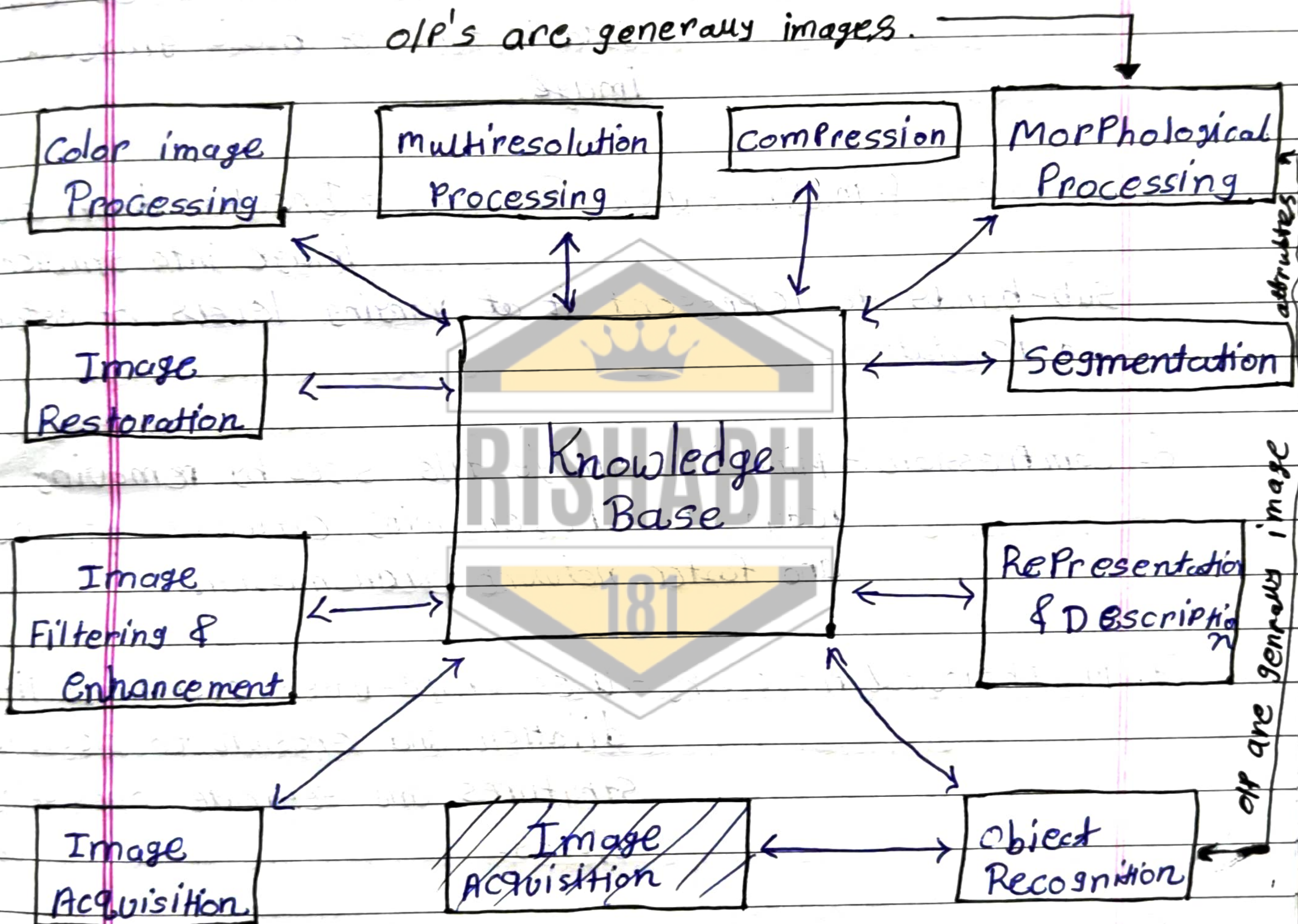
1. Note a block diagram of Components of digital image Processing in details.

⇒ Diagram :→



1. Image Acquisition :→ Captures a Physical Scence using a Sensor and Converts it Into a digital Pixel matrix.

2. Image Filtering and Enhancement :→ Sharpens details, adjusts contrast and filters noise to make an image look better or be more suitable for a Spefic task.

3. Image Restoration :→ uses mathematical models to reverse known degradation, such as fixing Camera blur, sensor noise or lens distortion.

4. Color Image Processing :→ Handles and manipulates color data instand of basic grayscale image.

5. Wavelets & mutirisoution Prossing :→ Breaks down an image into smaller sub-bands to represent it at varying levels of detail and resolution.

6. Compression :→ Minimizes image file sizes by removing redudant data, ensuring efficient strong and faster network transmission.

7. Morphological Processing :→ Uses Shape-based operations like dilation and erosion to clean up strutures and separate overlapping objects.

8. Segmentation :→ Cuts or partitions an image into distinct, meaningful regions.

9. Representation & Description :→ Converts raw segmented regions into a format the computer understands, extracting specific descriptors like shaps, texture or boundary metrics.

10. object Recongnition :→ Labels an object based on its extracted descriptors.

11. Knowledge Base :→ The central database that provides problem-specific rules, reference patterns and historical data ~~in~~ to guide every step of the workflow.

2. Explain image Samplings and image quantization.

⇒ Sampling :→ sampling is the process of converting a continuous image into a discrete image by dividing it into pixels.

→ Purpose :→ Determines the spatial resolution of an image.
:→ More samples produce higher image quality.

Ex :→ A 1024 X 1024 image ~~Process of ass~~ contains more detail than 256 × 256 image.

⇒ Quantization :→ Quantization is the process of assigning intensity values to sampled pixels.

→ Purpose :→ Determines the number of gray levels or color levels.
:→ Controls image brighness resolution.

Ex :→ 8-bit image → 256 gray levels (0-255)
1-bit image → 2 gray levels (Black and White)

3. Difference Between Image Sampling and Image Quantization.

	Image Sampling	Image Quantization
Defination →	Process of converting a continuous image into discrete Pixels.	Process of assigning discrete intensity values to sampled pixels.
Purpose →	Determines the spatial resolution of an image.	Determines the gray-level resolution of an image.
operation →	operates on the spatial Corrdinatess (x,y) of an image.	operates on the amplitude/ intensity values of Pixels.
mesurment unit →	measured in number of Pixels	measured in number of gray levels.
stage →	It is the first step in the image digitization Process.	It is the second step in the image digitization Process.
low value Result →	If too low, the image appears blocky or Pixelated.	If too low, the image shows false contours.

4. Explain type of image representation.

→ image Representation refers to the method used to store and reprensent images in a digital Computer system.

(1) Binary Image → A binary image contains only two Possible pixel values: 0 and 1.

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⇒ characteristics :→ 0 :→ Black
:→ 1 :→ white
:→ Requires only 1 bit per pixel
:→ Simplest image representation.

⇒ Applications :→ Document Processing
:→ Barcode scanning.

(2) Grayscale Image :→ A grayscale imge Consists of diffrent Shades of gray between black and white.

⇒ characteristics :→ Pixel values range from 0 to 255.
:→ 0 :→ Black
:→ 255 :→ white
:→ uses 8 bits per Pixel.

⇒ Applications :→ medical imaging
:→ Image enhancement

(3) Color Image :→ A color image represents images using Color information.

⇒ characteristics :→ uses RGB color model
:→ Each color channel uses 8 bits.
:→ Produces millions of color.

⇒ Applications :→ Digital Photography
:→ video Proessing
:→ multimedia applications